TRANSFORMING AMERICAN HEALTHCARE

Part II: The Six Pillars

Chapters 4–9

PART II: THE SIX PILLARS

CHAPTER 4: Policy — The Rules That Govern Everything

Why Policy Comes First

In the hierarchy of healthcare transformation, policy is not merely one variable among many. It is the operating environment within which every other variable functions. Clinical innovation that is not reimbursable under the policy framework cannot scale. Economic models that violate regulatory requirements cannot operate. Technology investments designed for a fee-for-service reimbursement world will be the wrong investments when that world changes.

This is why the Policy pillar, though not the most heavily weighted in the HTI (it ranks second at 20%), is analytically prior to the others. Before you can answer any question about economics, technology, clinical care, equity, or operations, you must first understand the policy rules within which those dimensions operate.

Policy analysis in healthcare has four major domains: federal legislation, federal administrative rules, state policy, and private market regulation. Each operates on different timescales, through different mechanisms, and with different implications for transformation.

The Federal Legislative Architecture

The Affordable Care Act (ACA) of 2010 remains the most structurally significant piece of healthcare legislation since Medicare and Medicaid were established in 1965. Understanding contemporary healthcare policy requires understanding what the ACA established, what it changed, and what it left unresolved.

What the ACA established:

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│               ACA STRUCTURAL ARCHITECTURE (2010)                 │
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│                                                                  │
│  COVERAGE EXPANSION                                              │
│  ├── Medicaid expansion to 138% FPL (optional for states)        │
│  ├── Insurance marketplaces (Exchanges) for individual market    │
│  ├── Premium tax credits for 100–400% FPL                        │
│  ├── Individual mandate (effectively eliminated 2019)            │
│  └── Dependent coverage to age 26                                │
│                                                                  │
│  INSURANCE MARKET REFORM                                         │
│  ├── Guaranteed issue: no denial for pre-existing conditions     │
│  ├── Community rating: age, geography, tobacco use only          │
│  ├── Essential Health Benefits: 10 categories required           │
│  ├── Medical Loss Ratio: 80% for small group, 85% for large      │
│  └── No lifetime or annual limits on essential benefits          │
│                                                                  │
│  PAYMENT REFORM INFRASTRUCTURE                                   │
│  ├── Center for Medicare and Medicaid Innovation (CMMI)          │
│  ├── Accountable Care Organization (ACO) program                 │
│  ├── Hospital Value-Based Purchasing Program                     │
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│   ├── Hospital Readmissions Reduction Program (HRRP)                │
│   └── Quality Payment Program (QPP) authorization                   │
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What the ACA changed permanently:

The ACA's insurance market reforms were not merely policy changes — they restructured the fundamental economics of the individual insurance market. Guaranteed issue (insurers must cover anyone regardless of health status) and community rating (premium variation limited to age, geography, and tobacco use) shifted the insurance market from a risk-selection industry to a risk-pooling industry. This shift, combined with premium tax credits and the mandate, was intended to create the large, balanced risk pools that make comprehensive individual coverage economically viable.

What the ACA left unresolved:

The ACA did not address the structural driver of cost: fee-for-service payment. It expanded coverage significantly and created infrastructure for payment reform. But the payment reform authority it created (CMMI) operates through voluntary models, not mandatory transformation. Fourteen years after the ACA, the majority of U.S. healthcare spending remains in fee-for-service.

CMMI: The Federal Laboratory for Payment Reform

The Center for Medicare and Medicaid Innovation (CMMI), established by the ACA, is the most important federal institution in healthcare transformation. CMMI has authority to test innovative payment and service delivery models without requiring new legislation, and to expand successful models nationwide if they reduce costs without reducing quality (or improve quality without increasing costs).

The CMMI model portfolio (selected, as of 2026):

Model	Type	Status	Key Finding
MSSP (Medicare Shared Savings Program)	Shared savings ACO	Ongoing, permanent program	~11 million beneficiaries; net savings ~$1.8B in 2023
ACO REACH	Full-risk global capitation	Active	Higher risk, higher potential savings than MSSP
Bundled Payments for Care Improvement (BPCI-A)	Episode bundled payment	Ongoing	Modest savings in orthopedic episodes
Comprehensive Primary Care Plus (CPC+)	Advanced primary care	Concluded	Mixed results; primary care investment without sufficient savings
Primary Care First (PCF)	Prospective primary care payment	Active	Enhanced primary care payment with quality performance
AHEAD Model (Vermont)	All-payer global budget	Active	Vermont-specific; most advanced model in portfolio
Making Care Primary (MCP)	Multi-payer primary care transformation	Active (launched 2024)	8-state initial cohort

The CMMI accountability problem:

CMMI's mandate is to test models that reduce costs and improve quality simultaneously. The challenge is that many models have produced modest or mixed results despite significant investment — leading to regular strategic reviews and model consolidations. The 2021 CMMI strategic refresh resulted in the retirement of several underperforming models and a renewed focus on accountable care models with genuine two-sided risk.

The core issue is that voluntary models attract a selection of organizations that are already performing better than average. A model that selects for high-performing ACOs will show savings compared to the FFS counterfactual — but those savings may reflect the selection of better organizations rather than the causal effect of the model itself. This methodological challenge is inherent to voluntary payment models and is one reason why advocates for mandatory global budgets argue that only universal program participation can produce reliable cost containment.

Medicare Payment Rules: The Annual Rulemaking Cycle

CMS publishes annual payment rules that update rates and policies for every major Medicare payment system. Understanding the annual rulemaking cycle is essential for healthcare financial planning:

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ANNUAL CMS RULEMAKING CALENDAR
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April/May:     Proposed IPPS Rule published (inpatient hospital)
               Proposed OPPS Rule published (outpatient hospital)
               Proposed Physician Fee Schedule (PFS) published

June/July:     Comment period closes

August:        Proposed rules finalized (hospitals)

October:       Proposed PFS finalized

November 1:    IPPS Final Rule takes effect (Fiscal Year start)

January 1:     PFS Final Rule takes effect (Calendar Year start)
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The annual rules set:

- **Base payment rates** for DRGs (inpatient), APCs (outpatient), and RVUs (physician)
- **Quality program requirements** for HVBP, HRRP, HAC Reduction Program
- **APM participation criteria** updates
- **Prior authorization policy** changes
- **Telehealth coverage** policy

For a hospital or physician group, failing to monitor and analyze these rules as they are proposed and finalized is a financial planning failure. A 2.3% market basket update to IPPS rates may look like a small annual adjustment — but if your payer mix is 45% Medicare, it is the single largest financial variable in your annual budget.

State Policy: The 50-State Laboratory

State governments have substantial authority over healthcare, particularly through their roles as:

1. **Medicaid program administrators** (with federal partnership)
2. **Insurance regulators** (for licensed commercial health plans)
3. **Hospital licensure and certificate of need (CON) authorities**
4. **Scope-of-practice regulators** (for health professions)

Medicaid waiver programs are the most important vehicle for state health policy innovation:

Section 1115 Waivers (Research and Demonstration Projects): Allow states to waive standard Medicaid requirements to test new approaches. Historically used for Medicaid expansion in non-ACA states, work requirements (mostly struck down), and delivery system reform initiatives (DSRIP programs). The key constraint: must be budget-neutral to the federal government.

Section 1115A Waivers (CMMI All-Payer Model Authority): The waiver authority used by Vermont's AHEAD Model. Allows CMMI to enter into agreements with states to test all-payer models that include Medicare, overriding standard Medicare payment rules for participating states. This is the most powerful state-level waiver authority — and the hardest to obtain.

Section 1332 Waivers (State Innovation Waivers): Allow states to waive ACA requirements for their individual and small-group insurance markets. Used for state reinsurance programs (Alaska, Maine, Minnesota, Colorado) that have successfully reduced premiums by 10–20% by removing high-cost enrollees from the individual market risk pool.

Prior Authorization: The Operational Policy Failure

Prior authorization (PA) — the requirement by health plans that providers obtain pre-approval before delivering certain services — is one of the most consequential and most contested policy issues in contemporary healthcare.

The problem with prior authorization:

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THE PRIOR AUTHORIZATION BURDEN (2024 DATA)
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Average PA requests per physician per week:  41
Hours per week spent on PA per physician:    13
Share of PA requests that are approved:      89%
Share that are eventually approved
  after denial and appeal:                   94%
Median PA wait time (specialty care):        14 days
Share of patients who abandon care
  while waiting for PA:                      27%

Annual cost of PA burden to U.S.
  healthcare system:                          ~$35 billion

Source: AMA Prior Authorization Survey, 2024
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The data reveals the fundamental policy paradox of prior authorization: the vast majority of requests are ultimately approved, yet the process consumes enormous clinician time, delays care for 27% of patients who abandon treatment while waiting, and costs the system tens of billions annually in administrative overhead — while not achieving its stated purpose of preventing inappropriate care in most cases.

The federal response:

The CMS Interoperability and Prior Authorization Final Rule (effective January 2024) requires health plans to:

- Implement electronic prior authorization (ePA) workflows
- Make PA decisions within 72 hours for urgent requests and 7 calendar days for standard requests
- Report PA approval and denial rates publicly

The state response:

More than 30 states have passed prior authorization reform legislation since 2020, requiring faster decisions, gold-carding (automatic approval for consistently approved providers), and independent external review. Vermont is among the states with the strongest PA reform requirements.

The gap between legislation and implementation: Despite these reforms, surveys indicate that PA burden has not materially declined, and that many health plans have found ways to technically comply with requirements while maintaining substantively unchanged processes. The 2024 CMS rule represents the most forceful federal intervention to date, but its full impact will not be measurable until 2026 data are available.

The Affordable Care Act's Unfinished Business

Sixteen years after the ACA, four structural gaps remain unaddressed:

1. The Medicaid expansion gap: Eleven states (as of 2026) have not adopted ACA Medicaid expansion, leaving an estimated 2.1 million people in a coverage gap — earning too much for traditional Medicaid but too little to qualify for marketplace subsidies. This gap falls disproportionately on low-income adults in Southern states with high rates of chronic disease.

2. The affordability gap for middle-income Americans: The ACA's premium tax credits phase out at 400% FPL (extended to 2025 by the ARP; subsequently extended further). Americans earning just above the subsidy cliff face premiums that can consume 15–20% of household income for comprehensive coverage. The ACA solved the affordability problem for low-income Americans; it did not solve it for middle-income Americans.

3. The fee-for-service persistence: As described above, despite fourteen years of payment reform activity, the majority of U.S. healthcare spending remains in fee-for-service. CMMI models have demonstrated that payment reform is possible; they have not demonstrated that it is scalable through voluntary participation alone.

4. The SDOH investment gap: The ACA did not create a sustainable funding mechanism for social determinants of health investment. Hospitals can screen patients for SDOH needs; they cannot pay for housing with hospital operating revenue. Closing the SDOH investment gap requires a funding architecture that the ACA did not provide.

Policy Analysis Tools: The HTR Policy Simulator

HTR's Research Lab includes a Policy Simulator that allows users to model the impact of policy changes across six scenario types:

1. Section 1115 Waiver Designer: Configure work requirements, premium assistance, benefit modifications, and managed care carve-outs. Models coverage impacts, enrollment changes, and state fiscal effects.

2. Global Budget Modeler: Set all-payer growth rate caps, quality risk corridors, and hospital-specific allocations. Models the fiscal implications of Vermont-style global budget mechanisms.

3. Medicaid Expansion Simulator: Estimates coverage gain by age and income band, state cost share under the enhanced federal match, and crowd-out effects on employer-sponsored insurance.

4. Price Transparency Policy: Estimates cost savings from shoppable service disclosure requirements under different market concentration scenarios. High-concentration markets (fewer provider choices) show lower transparency-driven savings than competitive markets.

5. Prior Authorization Reform: Models the administrative cost savings from PA burden reduction against the utilization cost impact of removing PA gatekeeping for different service categories.

6. State Benchmarking: Side-by-side comparison of two states' policy portfolios across all five HTI policy components.

Vermont as Policy Case Study: What One State's Legislative Agenda Reveals

Vermont's 2025–2026 legislative agenda illustrates how policy decisions interact across all six pillars simultaneously. The key policy questions on Vermont's legislative docket as of early 2026:

Reference-based pricing for hospital services (if enacted):

- *Policy:* Requires enabling legislation; may require federal waiver coordination
- *Economics:* Models suggest a reduction of 8–12% in commercial hospital revenue over 5 years
- *Technology:* Requires price transparency data infrastructure to be in place first
- *Clinical:* Risk of care access disruption if hospitals respond by reducing service lines
- *Equity:* Rural hospital revenue reductions disproportionate to cost structure
- *Operations:* Revenue cycle systems must be redesigned for rate-based rather than negotiated payment

This single policy proposal touches all six pillars. A legislature debating it needs a six-pillar analytical framework to assess it responsibly.

Key Concepts Introduced in This Chapter

> **Affordable Care Act (ACA):** The 2010 federal legislation that established the current architecture of U.S. health insurance coverage, introduced insurance market reforms, and created CMMI for payment reform.

> **CMMI:** The Center for Medicare and Medicaid Innovation, established by the ACA to test innovative payment and delivery models without new legislation.

> **Section 1115 waiver:** CMS authority allowing states to waive standard Medicaid requirements to test new approaches, subject to budget neutrality.

> **Prior authorization (PA):** A health plan requirement that providers obtain advance approval before delivering certain services; a major source of administrative burden and care delay.

> **DRG (Diagnosis-Related Group):** The payment classification system for inpatient hospital services under Medicare; hospitals receive a fixed payment per DRG regardless of actual costs incurred.

> **IPPS (Inpatient Prospective Payment System):** The Medicare payment system for inpatient hospital services, using DRGs updated annually by the CMS IPPS final rule.

> **Medical Loss Ratio (MLR):** The percentage of premium revenue a health plan spends on medical claims and quality improvement; ACA requires 80% for small group plans and 85% for large group plans.

CHAPTER 5: Economics — Following the Money Through Healthcare

The Central Thesis of Health Economics

Health economics begins with a premise that distinguishes healthcare from most other markets: **the normal mechanisms of market competition do not work well in healthcare**, and understanding why is the foundation for understanding everything else about healthcare finance.

In a standard competitive market, consumers choose between products based on price and quality information, competition among sellers drives prices toward costs, and the market self-corrects inefficiencies. In healthcare:

- **Patients cannot comparison-shop** for most services (emergencies are not shoppable; specialist care requires referrals; quality differences are not transparent)
- **Prices bear no systematic relationship to costs** (hospital charge-to-cost ratios average 300–400%, with enormous variation)
- **Insurers are intermediaries** between patients and providers, creating complex agency relationships that distort both price signals and utilization decisions
- **Information asymmetry is severe** (physicians know vastly more about the appropriate care than patients do, creating a principal-agent problem)
- **Externalities are significant** (vaccinations, infectious disease treatment, and mental health care have broad social benefits beyond the individual patient)

These market failures are why healthcare economics requires specialized analytical frameworks that go beyond standard microeconomic theory — and why payment reform (the policy attempt to fix these market failures) is so complex.

The Anatomy of U.S. Healthcare Spending

Where does the $4.5 trillion go?

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U.S. NATIONAL HEALTH EXPENDITURES — SPENDING BY CATEGORY (2023)
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Hospital care:                    $1,478B   (32.8%)
Physician & clinical services:    $1,029B   (22.9%)
Prescription drugs:                 $722B   (16.0%)
Nursing care & continuing care:     $191B    (4.2%)
Home health care:                   $161B    (3.6%)
Dental services:                    $152B    (3.4%)
Other health, residential, personal: $208B   (4.6%)
Government administration:           $316B   (7.0%)
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Investment (research, structures):   $243B   (5.4%)
─────────────────────────────────────────────────────────────────
TOTAL:                            $4,500B   (100%)

Source: CMS National Health Expenditure Accounts, 2023

Who pays:
  Private health insurance:  $1,397B   (31.0%)
  Medicare:                  $1,045B   (23.2%)
  Medicaid & CHIP:             $924B   (20.5%)
  Out-of-pocket:               $471B   (10.5%)
  Other government programs:   $437B    (9.7%)
  Other private (employers, etc.): $226B (5.0%)
```

The single most important feature of this spending profile: **administrative costs consume an estimated 25–34% of total healthcare spending** — between $1.1 trillion and $1.5 trillion annually — far more than any other comparator nation (Germany: ~14%, Canada: ~13%). This is not money that improves health. It is the cost of operating an extraordinarily complex, fragmented, multi-payer administrative apparatus.

Value-Based Care: The Economic Framework for Transformation

Value-based care (VBC) is the collective term for payment arrangements that tie provider reimbursement to quality, outcomes, or total cost of care rather than volume of services. It is the economic mechanism by which payment policy (Chapter 4) aims to realign incentives from volume to value.

The VBC spectrum ranges from relatively modest modifications to fee-for-service to complete risk transfer:

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THE VALUE-BASED CARE SPECTRUM
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                        RISK TO PROVIDER
                ◄──────────────────────────────────►
                LOW                              HIGH
                 │                                 │
                 ▼                                 ▼
      ┌────────────────┬─────────────┬────────────┬──────────────┐
      │  PAY FOR       │  BUNDLED    │  SHARED    │  GLOBAL      │
      │  PERFORMANCE   │  PAYMENTS   │  SAVINGS   │  CAPITATION  │
      │  (P4P)         │ (Episodes)  │  (MSSP)    │  (REACH)     │
      ├────────────────┼─────────────┼────────────┼──────────────┤
      │ FFS base +     │ Fixed price │ FFS base + │ Prospective  │
      │ bonuses or     │ per episode │ share of   │ per-member   │
      │ penalties for  │ (e.g., hip  │ savings vs.│ per-month    │
      │ quality        │ replacement)│ benchmark  │ payment      │
      ├────────────────┼─────────────┼────────────┼──────────────┤
      │ Small          │ Moderate    │ Moderate   │ Full         │
      │ incentive      │ downside    │ up/down    │ financial    │
      │ adjustment     │ possible    │ risk       │ risk         │
      └────────────────┴─────────────┴────────────┴──────────────┘
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Pay for Performance (P4P): The simplest VBC model. Providers receive their standard FFS payments plus or minus quality-based adjustments. CMS Hospital Value-Based Purchasing adjusts hospital Medicare payments by up to ±2% based on quality performance. P4P is nearly universal now but by itself has modest effects on total cost.

Bundled Payments: A single payment covers all services for a defined clinical episode — hip replacement, CABG, joint reconstruction — across all providers (hospital, surgeon, anesthesia, post-acute). The bundle holder (often the hospital) distributes payment internally. Creates strong incentives to reduce unnecessary post-acute utilization and hospital-acquired complications within the episode. Does not address non-episode utilization or population-level prevention.

Shared Savings (MSSP/ACO REACH): The most widely adopted two-sided risk model. Providers (organized as ACOs) take responsibility for the total cost of care for an attributed population. If they spend less than the benchmark, they share in the savings. If they spend more, they may owe back a portion (downside risk). The MSSP Basic track has limited downside risk; the Enhanced track and ACO REACH global option have full two-sided risk.

Global Capitation: The provider receives a fixed prospective payment per attributed member per month (PMPM), regardless of what services are actually delivered. The provider bears full financial risk for all healthcare spending within the defined scope. Vermont's AHEAD Model is a state-level global budget — structurally similar to capitation but applied across all payers rather than to a single insurer's population.

The MSSP Financial Model: A Deep Dive

The Medicare Shared Savings Program (MSSP) is the largest VBC program in the U.S. by covered lives. Understanding its financial mechanics illustrates the core economics of shared savings models.

Attribution: Beneficiaries are assigned to an ACO based on their primary care utilization — specifically, which primary care physicians they predominantly see. Attribution is retrospective (based on prior-year utilization patterns) or prospective (based on the prior year's primary care use to predict the current year's population).

Benchmark setting: The ACO's spending benchmark is set based on its attributed population's historical spending (risk-adjusted) blended with regional spending trends. The benchmark represents what CMS expects the ACO's population would have cost without the ACO.

Shared savings calculation:

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MSSP SHARED SAVINGS MECHANICS
───────────────────────────────
Step 1:  Benchmark = Risk-adjusted historical PMPM × Regional trend factor
Step 2:  Actual spend = Sum of all Part A/B FFS spending for attributed
population
Step 3:  Savings = Benchmark − Actual spend (if positive)
Step 4:  Apply Minimum Savings Rate (MSR) check
         (ACO must save at least 2–3.8% to keep any savings)
Step 5:  Gross savings × Sharing rate (50% for Basic; up to 75% Enhanced)
Step 6:  Apply Quality Performance Multiplier
         (ACO must meet quality threshold; higher quality = higher share)
Step 7:  Net shared savings payment to ACO

Example:
  Benchmark:           $1,200 PMPM × 10,000 lives = $144M
  Actual spend:        $1,140 PMPM × 10,000 lives = $136.8M
  Gross savings:       $7.2M
  MSR (3%):            $4.32M — threshold met
  Sharing rate (60%): $4.32M
  Quality multiplier: ×1.05 (high quality)
  Net payment:         $4.54M
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The MSR is critical: an ACO that saves $3.5 million on a $144 million benchmark (2.4%) keeps nothing under MSSP Basic Track. This is the design feature that creates the most frustration among ACO participants — meaningful savings that do not meet the statistical threshold for payment.

Hospital Financial Economics: Reading a Hospital Balance Sheet

For healthcare executives, understanding hospital financial performance is a foundational competency. The key metrics:

Operating margin:

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Operating Margin = (Total Operating Revenue − Total Operating Expenses)
                 / Total Operating Revenue × 100
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National median: +0.8% (2024). Vermont hospital median: -2.1%. A negative operating margin means the hospital is losing money on its core clinical operations — unsustainable without endowment draw or extraordinary items.

Days cash on hand:

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Days Cash on Hand = (Cash + Short-term Investments)
                  / (Total Operating Expenses / 365)
```

Measures how many days a hospital could operate with no revenue. Critical Access Hospital median: 87 days. Below 60 days signals financial stress. Below 30 days may trigger debt covenant violations.

Debt service coverage ratio:

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DSCR = Earnings Before Interest and Depreciation (EBIDA)
     / Annual Debt Service (Principal + Interest)
```

Lender covenant floor is typically 1.25×. Sustained DSCR below 1.0 means the hospital cannot service its debt from operations.

Payer mix and revenue concentration:

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VERMONT HOSPITAL PAYER MIX ANALYSIS (ESTIMATED 2024)
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Medicare (Traditional FFS):    44%  →  Lower reimbursement
Medicaid (FFS + MCO):          22%  →  Often below cost
Commercial (insured):          27%  →  Above cost; cross-subsidizes
Self-pay:                       3%  →  High bad debt risk
Other (workers' comp, etc.):    4%  →  Variable
───────────────────────────────────────────────────
Challenge: As commercial cross-subsidization is
constrained (by reference-based pricing proposals),
the financial model collapses without offsetting
public payment improvements.
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The cross-subsidization problem:

American hospitals survive financially by charging commercial payers far more than the cost of care — typically 250–400% of Medicare rates — to subsidize the losses they incur on Medicare and Medicaid patients. The Wyman Report's reference-based pricing recommendation (moving commercial hospital rates toward 200% of Medicare) would eliminate a significant portion of this cross-subsidy, accelerating the financial crisis in hospitals that have not built alternative revenue streams.

Actuarial Science: The Mathematics of Insurance Risk

Actuarial science underlies all health insurance pricing and is essential for understanding the economics of healthcare transformation. The core actuarial challenge: **predicting the future healthcare costs of a defined population so that premiums can be set appropriately**.

The actuarial value (AV) framework:

Under the ACA, individual market health plans are classified by metal tier based on their actuarial value — the percentage of total covered medical expenses the plan pays for a standard population:

Metal Tier	Actuarial Value	Consumer Pays
Bronze	60%	40%
Silver	70%	30%
Gold	80%	20%
Platinum	90%	10%

Actuarial value does not describe what any individual will pay — it describes what the plan pays on average across a standard population. A high-deductible bronze plan might pay nothing for a healthy person who avoids care but 60% for a chronically ill person who hits the deductible quickly.

Risk adjustment:

The ACA's permanent risk adjustment program transfers money from health plans with healthier-than-average populations to plans with sicker-than-average populations. This prevents the adverse selection death spiral that would occur if healthier people systematically enrolled in lower-cost plans, leaving sicker people concentrated in higher-cost plans.

The risk adjustment formula uses the HCC (Hierarchical Condition Category) model to score each enrollee's expected cost based on their diagnosis history, then calculates each plan's relative risk score. Plans above the average risk score receive transfers; plans below pay transfers.

Adverse selection and the death spiral:

The actuarial concept of adverse selection — the tendency for sicker people to buy more comprehensive insurance — is a structural challenge for individual insurance markets. Without mechanisms to counteract it (mandates, risk adjustment, guaranteed issue, reinsurance), markets tend toward a death spiral:

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INSURANCE MARKET DEATH SPIRAL
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Premiums rise → Healthy people drop coverage →
Pool gets sicker → Premiums rise further →
More healthy people leave → Pool gets sicker →
Eventually only sick people remain →
Market collapses
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The ACA's guaranteed issue, community rating, and risk adjustment were designed to prevent this spiral. They have largely succeeded in the individual market, though premium levels remain a significant barrier for middle-income Americans.

Cost-Effectiveness Analysis: Quantifying Value in Healthcare

Cost-effectiveness analysis (CEA) is the methodological framework for comparing the value of different healthcare interventions — answering the question: *How much does it cost to gain one additional unit of health outcome?*

The fundamental ratio:

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Incremental Cost-Effectiveness Ratio (ICER) =
  (Cost of Intervention - Cost of Comparator)
  ─────────────────────────────────────────────
  (Effectiveness of Intervention - Effectiveness of Comparator)
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Where effectiveness is typically measured in **QALYs** (Quality-Adjusted Life Years) — a unit that combines quantity of life (years lived) with quality of life (0 = death, 1 = perfect health).

Willingness-to-pay thresholds:

Organization	ICER Threshold	Notes
ICER (U.S.)	$100,000–$150,000/QALY	Most-cited U.S. benchmark
NICE (UK)	£20,000–£30,000/QALY	~$25,000–$37,000; much lower
WHO	1–3× GDP per capita	~$65,000–$195,000 in U.S.
CMS	No explicit threshold	Considers "reasonable and necessary"

What the ICER tells you: An intervention with an ICER below the threshold is considered cost-effective (society gets adequate value for the cost). Above the threshold, the intervention may still be valuable, but alternative uses of those resources would produce more health.

Vermont case study — CEA of behavioral health integration:

Vermont proposes to invest $40 million over 5 years in integrated behavioral health programs across 50 primary care sites. A hypothetical CEA:

- Incremental cost: $40M for the program, offset by ~$18M in reduced ED and inpatient utilization = net $22M incremental cost
- Effectiveness: estimated 2,400 QALYs gained from reduced mental health morbidity and mortality over the period
- ICER: $22M / 2,400 QALYs = $9,167/QALY — well below any standard threshold

This would make behavioral health integration one of the most cost-effective investments in Vermont's reform portfolio — which aligns with the evidence base and HTR's recommendation.

The Economics of Private Equity in Healthcare

Private equity (PE) investment in healthcare has grown dramatically since 2010, with PE-backed acquisitions of physician practices, nursing homes, behavioral health facilities, and hospitals accelerating through the mid-2020s. Understanding PE healthcare economics is essential for analysts, policymakers, and clinicians navigating this landscape.

How PE healthcare investments work:

PE firms acquire healthcare organizations using leveraged buyouts (LBOs) — financing acquisitions with large amounts of debt placed on the acquired entity. The strategy is to improve operational efficiency (often including cost-cutting), grow revenue (often through acquisition of competitors), and sell the portfolio company at a higher multiple within 5–7 years.

The quality and access concerns:

Research published in the New England Journal of Medicine, JAMA, and other journals has documented:

- Increased staffing shortages following PE acquisition of nursing homes
- Higher infection rates and higher mortality in PE-owned nursing homes
- Increased billing and coding intensity following PE acquisition of physician practices (driving up payer costs)
- Closure of unprofitable service lines (obstetrics, behavioral health, emergency departments) in PE-owned hospitals

The policy response:

States including New York, California, and Massachusetts have enacted legislation requiring AG review of healthcare acquisitions and imposing conditions on PE-backed transactions. Federal antitrust enforcement has increased scrutiny of horizontal hospital consolidation, though vertical consolidation (insurer-provider integration) has received less consistent scrutiny.

Key Concepts Introduced in This Chapter

> **Value-based care (VBC):** Any payment arrangement tying reimbursement to quality, outcomes, or total cost of care rather than volume.

> **Medicare Shared Savings Program (MSSP):** The largest ACO program in the U.S.; ACOs share in savings (and potentially losses) against a risk-adjusted benchmark.

> **ACO REACH:** The successor to the Global and Professional Direct Contracting models; a full-risk ACO program with prospective capitation options.

> **Actuarial value (AV):** The percentage of total expected medical costs covered by a health plan for a standard population; the basis for ACA metal tier classification.

> **ICER (Incremental Cost-Effectiveness Ratio):** The ratio of incremental cost to incremental effectiveness between two interventions; the primary metric in cost-effectiveness analysis.

> **QALY (Quality-Adjusted Life Year):** The standard effectiveness unit in cost-effectiveness analysis; one year of life in perfect health.

> **Adverse selection:** The tendency of higher-risk individuals to disproportionately seek insurance, potentially destabilizing insurance markets.

> **HCC (Hierarchical Condition Category):** The risk-adjustment model used in Medicare, Medicare Advantage, and the ACA to adjust payments based on enrollees' predicted healthcare costs.

CHAPTER 6: Technology — Digital Infrastructure for the New Healthcare

The Technology Paradox

Healthcare technology presents a paradox: the sector that most needs information technology to function — that literally makes life-and-death decisions based on information — is among the latest industries to adopt standardized digital information exchange. As recently as 2009, fewer than 10% of U.S. hospitals used basic electronic health records.

The reasons for this late adoption are structural. Healthcare data is extraordinarily sensitive (privacy concerns created regulatory barriers to sharing). The clinical workflow requirements for documentation are complex and highly variable. EHR vendors built proprietary systems designed to retain data — and customers — within their platforms. The absence of a single national healthcare information network (unlike, say, the banking system's real-time interbank settlement) meant that interoperability was never operationally required.

The HITECH Act (2009) and the ACA (2010) changed this by funding EHR adoption and establishing federal standards for interoperability. But funding adoption and achieving interoperability are different things. By 2020, nearly all U.S. hospitals had EHR systems — but most of those systems could not exchange data with each other in a clinically meaningful way.

FHIR: The Standard That Is Transforming Healthcare Data Exchange

Fast Healthcare Interoperability Resources (FHIR) is the HL7 standard for representing and exchanging healthcare information electronically. Developed beginning around 2011 and first published as a formal standard in 2019 (R4), FHIR represents a fundamental architectural shift from prior healthcare data standards.

Pre-FHIR: the HL7 v2 era

The dominant standard before FHIR was HL7 version 2 (v2), developed in the late 1980s. HL7 v2 is a pipe-delimited message format used to transmit clinical events (admissions, lab results, orders) between hospital systems. It remains widely used today — the installed base is enormous — but it has fundamental limitations:

- Every v2 implementation is slightly different; "standard" messages require custom mapping
- It is a point-to-point messaging system, not an API-based query system

- It does not support the query patterns required for population health management
- Security and access control are external to the standard

FHIR: what it changes

FHIR defines healthcare information as a set of modular **resources** — discrete data objects representing clinical and administrative concepts:

```
KEY FHIR R4 RESOURCES
──────────────────────
Patient         →  Demographic information
Practitioner    →  Provider information
Organization    →  Health organization
Condition       →  Diagnoses (ICD-10 codes)
Observation     →  Lab results, vital signs (LOINC codes)
MedicationRequest → Prescription orders
Procedure       →  Completed procedures (CPT/SNOMED)
Encounter       →  Clinical encounters (visits, admissions)
Claim           →  Insurance claims
Coverage        →  Insurance coverage details
CarePlan        →  Care management plans
AllergyIntolerance → Allergy records
```

Each resource is a JSON or XML document with a defined structure. Resources reference each other (a MedicationRequest references a Patient and a Practitioner), creating a linked data model.

FHIR's **RESTful API** allows queries like:

- `GET /Patient/{id}` — retrieve a patient record
- `GET /Condition?patient={id}&code=E11` — retrieve all Type 2 diabetes diagnoses for a patient
- `GET /MedicationRequest?patient={id}&status=active` — retrieve active prescriptions

This API-based architecture makes FHIR far more compatible with modern application development patterns and enables the population-level queries that value-based care requires.

The 21st Century Cures Act and FHIR Mandates

The 21st Century Cures Act (2016) and its implementing regulations (ONC Information Blocking Final Rule, 2020; CMS Interoperability and Patient Access Final Rule, 2020) created the most significant federal mandates for health data interoperability in history.

Key requirements:

Patient Access API (mandatory for CMS-regulated payers): All Medicare Advantage, Medicaid, CHIP, and ACA marketplace plans must implement a FHIR R4 Patient Access API allowing patients to access their claims data, encounter data, and clinical data via third-party applications of their choosing.

Provider Access API (phased in through 2027): Payers must implement APIs allowing providers to access their patients' clinical histories across payers — critical for care coordination and value-based care management.

Payer-to-Payer API: When a patient changes insurance, the new payer must request and receive the patient's clinical data from the prior payer via FHIR API.

Information Blocking Rule: The 21st Century Cures Act prohibits "information blocking" — practices by EHR vendors, health systems, or payers that interfere with the access, exchange, or use of electronic health information. Violations can result in substantial civil monetary penalties.

ONC HealthIT Certification: EHR systems must be certified to ONC requirements including FHIR R4 API support.

EHR Systems: The Current Landscape

Electronic Health Records are the central nervous system of clinical operations and the primary source of clinical data for value-based care performance. Understanding the EHR landscape is essential for any healthcare technology analysis.

Market concentration:

The EHR market is highly consolidated. Three vendors — Epic, Oracle Health (formerly Cerner), and Meditech — account for approximately 75–80% of inpatient hospital market share. In the ambulatory market, Epic and Veeva dominate large practices; AthenaHealth, eClinicalWorks, and Allscripts serve smaller practices.

Epic's market share has grown significantly through the 2020s, partly because its Care Everywhere network — an Epic-proprietary interoperability layer — enables data sharing among Epic users while FHIR-based cross-EHR interoperability remains inconsistent in practice.

The interoperability gap:

Despite FHIR mandates, true clinical interoperability — the ability for a physician to see a complete, structured clinical history for a patient regardless of where prior care was received — remains unrealized in most clinical settings as of 2026. The reasons:

1. **Data quality:** FHIR APIs expose data; they do not guarantee data quality. A patient's problem list in one EHR may be outdated, incomplete, or differently coded than in another.

2. **Patient matching:** Without a universal patient identifier (prohibited by law in the U.S. since 1998), matching records across systems relies on probabilistic algorithms using name, date of birth, and address — which fail at rates of 5–10%.

3. **Workflow integration:** Receiving data via FHIR API is different from having that data integrated into the clinician's workflow. Most clinicians do not proactively query external FHIR APIs; the data must appear in their native EHR workflow automatically.

4. **Vendor economic incentives:** EHR vendors have historically competed on data lock-in. While the Information Blocking Rule prohibits explicit barriers, vendors still compete on the depth and quality of their internal analytics, which creates implicit incentives against deep interoperability.

Artificial Intelligence in Healthcare: Promises and Perils

Artificial intelligence is reshaping healthcare at every level — from diagnostic imaging interpretation to natural language processing of clinical notes to predictive analytics for population health management. Understanding AI's capabilities and risks is essential for every healthcare professional.

AI capability domains in healthcare:

1. Diagnostic AI: Computer vision models applied to radiology images (chest X-rays, CT scans, MRI, pathology slides) can detect findings at or above radiologist accuracy for specific conditions. FDA-cleared AI diagnostic devices number in the hundreds as of 2026, with applications spanning radiology, ophthalmology (diabetic retinopathy screening), dermatology, and cardiology (ECG interpretation).

2. Predictive analytics: Machine learning models applied to EHR and claims data to predict:

- Sepsis onset (2–12 hours before clinical recognition)
- 30-day hospital readmission risk
- Emergency department utilization
- Chronic disease progression (diabetes complications, heart failure exacerbation)
- Care gap identification (patients due for preventive services)

3. Clinical decision support (CDS): AI-powered alerts and recommendations delivered within the EHR workflow at the point of care. CDS Hooks is the FHIR-based standard for CDS integration — it allows third-party CDS services to receive context from the EHR (current patient, current order) and return recommendations (drug interaction alerts, guideline adherence reminders, care gap flags).

4. Natural language processing (NLP): Extracting structured information from unstructured clinical notes — diagnoses, medications, symptoms, social history — to support quality measurement, risk stratification, and research without requiring structured documentation.

5. Administrative AI: Prior authorization automation, medical coding, denial prediction, and patient scheduling optimization. Administrative AI represents some of the highest-ROI near-term applications of AI in healthcare because the administrative burden is enormous and well-defined.

Algorithmic Bias: The Equity Risk in Healthcare AI

Algorithmic bias in healthcare AI is not a hypothetical concern — it is a documented, consequential problem with specific mechanisms and real-world examples.

Mechanism 1: Biased training data

A predictive model trained on historical clinical data will encode the patterns in that data. If Black patients historically received less aggressive treatment for a given condition than white patients (documented in the literature for many conditions), a model predicting "appropriate care" based on historical patterns will recommend less aggressive treatment for Black patients — not because race is a direct input, but because historical patterns encode racial disparities.

Mechanism 2: Proxy variables

Race is often not directly used in clinical AI models — but many variables that are used are correlated with race (zip code, insurance type, prior utilization patterns). A model that excludes race may still encode racial differences through these proxy variables.

Documented example — the Obermeyer et al. study (2019):

A widely used commercial algorithm to identify patients for high-risk care management used healthcare costs as a proxy for health needs. Because Black patients historically had lower healthcare costs than white patients with the same underlying health conditions — due to reduced access to care, not better health — the algorithm systematically underidentified Black patients as high-risk and undertargeted them for care management programs. The study estimated the algorithm reduced Black patients' enrollment in care management by approximately 50%.

Bias assessment framework:

HTR's AI Analytics Lab evaluates clinical AI tools on three bias metrics:

```
ALGORITHMIC BIAS METRICS
─────────────────────────────────────────────────────────
Demographic Parity Difference:
  Difference in positive prediction rate between groups
  Acceptable threshold: <0.05 (5 percentage points)

Equal Opportunity Difference:
  Difference in true positive rate (sensitivity) between groups
  Acceptable threshold: <0.05

Disparate Impact Ratio:
  Ratio of positive prediction rates between groups
  Acceptable threshold: >0.80 (80% rule from employment law)
─────────────────────────────────────────────────────────
A model passing all three tests is not guaranteed equitable;
it is a minimum bar, not a certification.
```

Telehealth: Policy, Technology, and Equity at the Intersection

Telehealth — the delivery of healthcare services via audio-video (or audio-only) technology — was the most rapidly adopted healthcare technology innovation of the COVID-19 pandemic era. Understanding telehealth's policy, technology, and equity dimensions illustrates the six-pillar interactions clearly.

Telehealth penetration trajectory:

```
TELEHEALTH UTILIZATION — SHARE OF OUTPATIENT VISITS
───────────────────────────────────────────────────
Pre-COVID (Q1 2020):       <1% of outpatient visits
Peak COVID (Q2 2020):      ~52% of outpatient visits
Post-PHE stabilization:    ~15–18% of outpatient visits (2025)
Rural: higher penetration; urban specialty: lower
```

Policy dimensions:

- Telehealth waivers during the COVID-19 Public Health Emergency (PHE) expanded coverage to include audio-only visits, home location origination, and new service types
- Congress has repeatedly extended telehealth flexibilities; permanent authorization remains a legislative priority
- State licensure laws require physicians to be licensed in the state where the patient is located — creating a patchwork of multi-state licensing requirements

Technology dimensions:

- Platform interoperability: most telehealth platforms do not integrate with EHR systems, creating documentation fragmentation
- FHIR-based telehealth integration is an emerging standard but not yet widely deployed
- Audio-only telehealth (phone calls) remains important for populations without broadband or video capability

Equity dimensions:

- Telehealth can improve access for rural patients (eliminating travel to specialists)
- But broadband access gaps mean rural and low-income patients are less likely to have reliable video capability
- Digital literacy barriers disproportionately affect elderly and low-income patients
- Audio-only telehealth restores some equity but is reimbursed at lower rates by CMS

The FHIR Lab: Building and Validating FHIR Resources

HTR's FHIR Interoperability Lab provides hands-on experience with FHIR R4 resource construction and validation. Here is an illustrative example of a FHIR R4 Patient resource:

```
{
  "resourceType": "Patient",
  "id": "vt-patient-example-001",
  "meta": {
    "profile": ["http://hl7.org/fhir/us/core/StructureDefinition/us-core-patient"]
  },
  "identifier": [{
    "use": "official",
    "type": {
      "coding": [{"system": "http://terminology.hl7.org/CodeSystem/v2-0203", "code": "MR"}]
    },
    "value": "12345-VT"
  }],
  "name": [{"use": "official", "family": "Doe", "given": ["Jane"]}],
  "gender": "female",
  "birthDate": "1958-03-15",
```

```
    "address": [{
      "use": "home",
      "state": "VT",
      "postalCode": "05401"
    }],
    "extension": [{
      "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-race",
      "extension": [{"url": "ombCategory", "valueCoding": {
        "system": "urn:oid:2.16.840.1.113883.6.238",
        "code": "2054-5",
        "display": "Black or African American"
      }}]
    }]
  }
```


The US Core extension for race/ethnicity illustrates a key equity-technology intersection: collecting structured race/ethnicity data in a standardized, machine-readable format is a prerequisite for algorithmic bias testing and health equity measurement. When race data is missing or inconsistently coded, equity analysis becomes impossible.

Key Concepts Introduced in This Chapter

> **FHIR (Fast Healthcare Interoperability Resources):** The HL7 R4 standard for representing and exchanging healthcare data via RESTful APIs; the foundation of modern health data interoperability.

> **21st Century Cures Act:** 2016 federal legislation establishing patient data access rights and prohibiting information blocking; implemented via ONC and CMS rules through 2020–2024.

> **Patient Access API:** FHIR-based API required of CMS-regulated payers, giving patients access to their claims and clinical data via third-party applications.

> **CDS Hooks:** The FHIR-based standard for integrating clinical decision support into EHR workflows at the point of care.

> **HCC v28:** The 2024 update to the Hierarchical Condition Category risk-adjustment model; used by Medicare Advantage, MSSP, and ACO REACH for risk scoring.

> **Algorithmic bias:** Systematic, measurable disparities in AI model performance across demographic groups; a significant equity and patient safety concern in clinical AI deployment.

> **Information blocking:** Practices that interfere with the access, exchange, or use of electronic health information; prohibited under the 21st Century Cures Act with civil monetary penalties.

CHAPTER 7: Clinical — Where Transformation Meets the Patient

The Purpose of the Clinical Pillar

All the policy reform, economic restructuring, and technology investment described in the preceding chapters serves exactly one ultimate purpose: changing what happens when a patient and a clinician meet. The Clinical pillar is where transformation is validated or exposed as insufficient.

This chapter covers four domains of clinical transformation: care delivery model redesign, clinical quality measurement, workforce dynamics, and the specific clinical reforms most relevant to healthcare transformation. Each domain is both a subject of knowledge and a set of analytical questions that practitioners must be able to answer.

Care Delivery Model Redesign

The Patient-Centered Medical Home

The **Patient-Centered Medical Home (PCMH)** is a team-based primary care model recognized by NCQA through its PCMH Recognition Program. The PCMH model organizes primary care around five core functions:

```
PCMH FIVE CORE FUNCTIONS
─────────────────────────────────────────────────────
1. COMPREHENSIVE CARE
   Team-based care with behavioral health, pharmacy,
   and social work integrated in primary care setting

2. PATIENT-CENTERED CARE
   Whole-person care, patient preferences honored,
   longitudinal relationship with care team

3. COORDINATED CARE
   Care coordination across specialty, hospital,
   home, and community settings; transition management

4. ACCESSIBLE SERVICES
   Enhanced access: same-day appointments, after-hours
   care, 24/7 clinical advice, telehealth

5. QUALITY AND SAFETY
   Quality improvement, evidence-based care,
   performance measurement and reporting
─────────────────────────────────────────────────────
```

PCMH has strong evidence for reducing ED utilization, improving preventive care rates, and increasing patient experience scores. The financial model challenge: PCMH investment requires upfront costs (care coordination staff, technology, workflow redesign) whose returns come primarily through reduced downstream utilization — savings that accrue to payers, not necessarily to the primary care practice. PCMH succeeds financially only within VBC payment models that share savings with the practice.

Accountable Care Organizations

An **Accountable Care Organization (ACO)** is a network of providers — typically organized around a hospital or large physician group — that collectively takes accountability for the cost and quality of care for a defined population under a value-based contract. ACOs are the primary mechanism through which MSSP and ACO REACH operate.

The defining characteristic of an ACO is **shared accountability**: the ACO's performance — both financially and on quality measures — is assessed at the network level, not the individual provider level. This creates incentives for care coordination across the network and for investment in population health programs that reduce costs systemically.

ACO operational requirements:

- **Attribution model:** Define and track the attributed population throughout the performance year
- **Care management program:** Identify high-risk patients, engage care managers, and track care management touchpoints
- **Quality measurement:** Report HEDIS, CMS Stars, and ACO-specific quality measures
- **Data analytics:** Real-time claims analysis, care gap identification, risk stratification
- **Network management:** Ensure specialist and post-acute partners understand and support ACO goals

Integrated Behavioral Health

Behavioral health (mental health and substance use disorder) is the most underinvested domain in American healthcare relative to its disease burden. Mental health conditions and substance use disorders:

- Affect approximately 52 million adults annually (1 in 5)

- Are present as a comorbidity in >70% of high-cost, high-utilization patients
- Account for approximately $280 billion in annual healthcare spending
- Generate massive indirect costs through lost productivity, incarceration, and homelessness

The **Collaborative Care Model (CoCM)** — integrating a psychiatric consultant and a behavioral health care manager into the primary care team — has the strongest evidence base for improving outcomes in primary care behavioral health integration. CMS reimburses CoCM services under CPT codes 99492–99494.

Vermont's integrated behavioral health investment — a key recommendation of both the Oliver Wyman report and HTR's advisory analysis — targets this high-leverage clinical intervention.

Clinical Quality Measurement: The HEDIS, Stars, and MIPS Framework

HEDIS: The Foundation of Health Plan Quality Measurement

HEDIS (Healthcare Effectiveness Data and Information Set) is maintained by NCQA and consists of 90+ measures spanning six domains:

Domain	Examples
Effectiveness of Care	HbA1c control for diabetics, colorectal cancer screening, breast cancer screening
Access & Availability	Adults' access to preventive/ambulatory care, children's access to PCPs
Experience of Care	CAHPS survey results — communication, care coordination
Utilization & Risk	Antibiotic utilization, opioid prescribing rates
Health Plan Descriptive	Disenrollment, mental health parity compliance
Equity	Stratification of HEDIS measures by race/ethnicity

HEDIS data is collected through two methods:

- **Administrative data:** Claims data submitted to NCQA — used for measures with complete claims capture
- **Hybrid method:** Random sample of administrative records supplemented by chart review — used when claims data is insufficient

Key HEDIS measures for value-based care:

```
HIGH-WEIGHT HEDIS MEASURES FOR VBC
────────────────────────────────────────────────────────────────
CDC — Diabetes Care: HbA1c Testing           (poor control: >9%)
CDC — Diabetes Care: Blood Pressure Control  (<140/90)
CBP — Controlling High Blood Pressure        (<140/90)
BCS — Breast Cancer Screening               (mammography, 50–74)
COL — Colorectal Cancer Screening           (50–75)
FUH — Follow-Up After Mental Health Hosp.  (7-day, 30-day)
SAA — Adherence to Antipsychotics           (schizophrenia)
FPC — Follow-Up After ED for Mental Illness (7-day)
AMM — Antidepressant Medication Management  (acute, continuation)
TRC — Transitions of Care                   (post-inpatient)
────────────────────────────────────────────────────────────────
Each measure is benchmarked to NCQA national averages.
75th percentile performance is the standard quality threshold
for most value-based contracts.
```

CMS Star Ratings

CMS Star Ratings apply to Medicare Advantage (MA) and Part D plans, rating them on a 1–5 star scale based on a composite of 40+ measures including HEDIS-based clinical quality, patient experience (CAHPS), and administrative performance (call center, appeals).

Why Stars matter financially: MA plans rated 4 or more stars receive a **quality bonus payment** of 5% above the standard CMS benchmark rate per enrolled beneficiary — plus additional bonus for 5-star plans. On a plan with 100,000 enrollees and an average monthly benchmark of $1,000, the difference between 3.5 and 4.0 stars is approximately $60 million annually in additional federal funding.

Star Rating sensitivity analysis:

The clinical quality domain drives the largest portion of the Star Rating composite. The most commonly missed stars involve:

- Medication adherence measures (requiring active member outreach)
- HbA1c control (requiring intensive diabetes management)
- Annual flu vaccines (requiring proactive outreach campaigns)
- Mental health follow-up (requiring care transition coordination)

An MA plan that analyzes its star measure performance and prioritizes the measures where it is closest to the next star cutpoint — rather than across-the-board quality improvement — will maximize its ROI on quality investment.

MIPS: Quality Payment for Physicians

The **Merit-based Incentive Payment System (MIPS)** governs quality-based adjustments to Medicare physician payments. MIPS has four performance categories:

Category	Weight	What It Measures
Quality	30%	Performance on 6 selected HEDIS/NQF measures
Cost	30%	Total Per Capita Cost and Medicare Spending Per Beneficiary
Improvement Activities	15%	Care coordination, patient engagement, PCMH recognition
Promoting Interoperability	25%	EHR use, FHIR API activation, immunization registry

Payment adjustment range: ±9% of Medicare Part B payments based on composite MIPS score.

MIPS creates a complex optimization problem: which combination of quality measures, improvement activities, and interoperability reporting maximizes the MIPS composite score given the practice's patient population and existing infrastructure? HTR's Clinical Quality Optimizer models this optimization systematically.

Risk Stratification: Identifying Who Needs What Care

Risk stratification — the process of identifying which patients are at highest risk of costly health events — is the analytical foundation of population health management. You cannot invest care management resources equitably and efficiently without first knowing which patients need them most.

The HCC Risk Adjustment Framework:

The Hierarchical Condition Category (HCC) model, used by CMS for Medicare Advantage risk adjustment, assigns each patient a **Risk Adjustment Factor (RAF) score** based on their age, gender, and diagnosis history. The RAF score predicts what the patient will cost in the next year relative to the average Medicare beneficiary (RAF = 1.0).

HCC v28 structure:

```
RAF SCORE CONSTRUCTION
──────────────────────────────────────────────────────────────
Demographic component:
  Age–sex coefficient (varies by age/gender)

Disease component:
  Sum of condition coefficients for all qualifying HCC codes

Interaction terms:
  Additional coefficients for condition combinations
```

```
  (e.g., diabetes + congestive heart failure)

Example — 72-year-old male with:
  HCC 19 (Diabetes without complications):     +0.118
  HCC 85 (Congestive heart failure):           +0.323
  HCC 18 + 85 interaction (DM + CHF):          +0.156
  Age-sex coefficient (M, 72):                  +0.447
  ─────────────────────────────────────────────────────
  RAF Score:                                    1.044

Interpretation: This patient is expected to cost 4.4%
more than the average Medicare beneficiary.
```

Population segmentation for care management:

```
RISK TIER FRAMEWORK
────────────────────────────────────────────────────────────────
TIER 1: COMPLEX (RAF ≥ 3.0)        ~3% of population, ~40% of cost
  → Intensive case management, dedicated nurse case manager,
    monthly touchpoints, high-touch care transitions

TIER 2: HIGH RISK (RAF 1.5–2.9)    ~12% of population, ~30% of cost
  → Care management program, quarterly PCP engagement,
    care coordination, medication management

TIER 3: RISING RISK (RAF 0.8–1.4)  ~25% of population, ~20% of cost
  → Disease management, preventive care outreach,
    chronic disease education

TIER 4: LOW RISK (RAF < 0.8)       ~60% of population, ~10% of cost
  → Preventive care, annual wellness visits, population-level
    health education
────────────────────────────────────────────────────────────────
```

Workforce: The Human Constraint on Transformation

Every clinical transformation model depends on human beings to deliver it. The workforce available to implement transformation is finite, geographically distributed, and under severe strain. Understanding workforce supply and demand is not background knowledge — it is a binding constraint on what transformation is possible.

National workforce projections (HRSA, 2024):

Workforce Category	2024 Supply	2035 Demand	Projected Gap
Primary care physicians	250,000	318,000	-68,000
Registered nurses	3.1 million	3.6 million	-500,000
Mental health counselors	280,000	395,000	-115,000
Community health workers	59,000	120,000	-61,000
Home health aides	3.6 million	4.8 million	-1.2 million

Workforce economics:

```
NURSING TURNOVER COST ANALYSIS (2024)
──────────────────────────────────────
Average RN salary:                $84,000/year
Turnover cost as % of salary:     75–130%
Per-nurse replacement cost:       $63,000–$109,000
```

```
Annual national RN turnover rate: ~22%
Total annual cost of RN turnover: ~$5.5B nationwide
```

These numbers explain why workforce retention is not merely an HR concern — it is a financial and operational imperative.

The scope-of-practice policy lever:

One of the fastest near-term responses to primary care workforce shortages is expanding the scope of practice for Advanced Practice Registered Nurses (APRNs — Nurse Practitioners and Certified Nurse Midwives) and Physician Assistants (PAs).

APRN full practice authority (allowing APRNs to practice without physician supervision) is granted in 26 states as of 2026. Studies consistently show that APRN care quality is equivalent to physician care for primary care services, and that APRN practice in underserved areas increases access to care significantly.

Vermont's scope-of-practice expansion for APRNs is one of HTR's highest-priority endorsed recommendations — the fastest path to primary care capacity expansion in a state with rural access challenges.

Preventable Hospitalizations: The Clinical Measure of System Failure

Prevention Quality Indicators (PQIs) — AHRQ's set of measures identifying hospitalizations that should not occur with adequate ambulatory care — are among the most important clinical quality metrics from a transformation perspective. Each PQI hospitalization represents a clinical and system failure: a condition that appropriate outpatient care should have prevented.

High-priority PQIs:

PQI	Condition	Cost per Event	System Meaning
PQI 1	Diabetes short-term complications	~$8,500	Inadequate diabetes primary care
PQI 5	COPD or asthma (adult)	~$9,200	Inadequate pulmonary management
PQI 7	Hypertension	~$7,400	Blood pressure uncontrolled
PQI 8	CHF	~$12,100	Inadequate heart failure management
PQI 11	Community-acquired pneumonia	~$11,800	Inadequate preventive care
PQI 14	Uncontrolled diabetes	~$8,800	Inadequate diabetes management
PQI 15	Asthma (adult)	~$7,900	Inadequate asthma management

Vermont's rural access challenges manifest directly in PQI rates: rural Vermonters have measurably higher rates of preventable hospitalizations for diabetes complications, COPD, and CHF — driven by limited primary care access and the social isolation that prevents timely care-seeking.

Key Concepts Introduced in This Chapter

> **PCMH (Patient-Centered Medical Home):** NCQA-recognized primary care model emphasizing team-based, coordinated, accessible, and patient-centered care.

> **HEDIS:** Healthcare Effectiveness Data and Information Set; 90+ quality measures maintained by NCQA for evaluating health plan and provider performance.

> **HCC (Hierarchical Condition Category):** Risk-adjustment model that scores patients' expected future healthcare costs based on age, sex, and diagnosed conditions.

> **RAF (Risk Adjustment Factor):** The HCC-derived score predicting a patient's cost relative to the average Medicare beneficiary; used in Medicare Advantage and ACO payment models.

> **Preventable hospitalization (PQI):** An AHRQ Prevention Quality Indicator hospitalization — a hospital admission for a condition that adequate ambulatory care should have prevented.

> **Collaborative Care Model (CoCM):** An evidence-based model for integrating behavioral health into primary care via a psychiatric consultant and behavioral health care manager.

> **MIPS:** Merit-based Incentive Payment System; CMS quality-payment program adjusting physician Medicare payments by up to ±9% based on quality, cost, interoperability, and improvement activities.

CHAPTER 8: Health Equity — The Cross-Cutting Imperative

Why Equity Is a Pillar, Not an Afterthought

Health equity analysis has often been treated as a supplement to the "real" analysis — a section added to policy documents, a diversity slide added to presentations, a sensitivity run added to economic models. This framing is wrong analytically and dangerous practically.

Health equity is not a niche concern. It is central to every transformation objective:

- **Cost containment** efforts that ignore disparities will reduce costs for healthier populations while leaving high-cost, high-complexity populations (who are disproportionately non-white and low-income) underserved — and financially unsustainable.
- **Quality improvement** programs that measure average performance will miss the fact that average performance can coexist with severe disparate outcomes by subgroup.
- **Technology** that trains on biased data will amplify disparities at scale, not reduce them.
- **Workforce** shortages fall hardest on rural and underserved communities where disparities are greatest.
- **Operations** that do not design for language access, cultural competence, and multi-modal outreach will systematically underserve the populations with the highest needs.

The equity pillar exists not to add a sixth analysis to a completed five-pillar framework. It exists to ensure that the other five pillars are designed and evaluated in ways that work for everyone — not just for the average patient.

The Anatomy of Health Disparities

Health disparities are measurable, systematic differences in health outcomes between population groups that are socially produced — not random, not inevitable, and not explained by biological differences between groups.

Racial and ethnic disparities: documented magnitude

```
SELECTED RACIAL/ETHNIC HEALTH DISPARITIES (U.S., 2024)
──────────────────────────────────────────────────────────
Measure                      White    Black    Hispanic  Asian
─────────────────────────────────────────────────────────────
Life expectancy (years)       78.3     72.8       77.9    84.1
Infant mortality (per 1,000)   4.3      10.6       4.8     3.6
Maternal mortality (per 100K) 13.4     37.1       12.3    6.0
Hypertension control rate (%)  54%      45%        53%    57%
Diabetes prevalence (%)         9%      12%        13%     9%
Uninsured rate (%)              7%       9%        18%     6%
Depression tx. receipt rate (%) 58%     34%        28%    21%
──────────────────────────────────────────────────────────────
Source: CDC, AHRQ National Healthcare Quality and Disparities Report, 2024
```

The Black maternal mortality rate — 37.1 per 100,000 live births, nearly 3× the rate for white women — is among the most widely cited examples of systemic racism in healthcare. This disparity exists at every income level and educational attainment level; it is not explained by socioeconomic status.

The geographic dimension:

```
RURAL vs. URBAN HEALTH DISPARITIES
────────────────────────────────────────────────────────────────
Premature death rate (before 75):    Rural 32% higher than urban
Cancer death rate:                   Rural 18% higher
Heart disease death rate:            Rural 24% higher
Unintentional injury death rate:     Rural 50% higher (incl. opioids)
Suicide rate:                        Rural 45% higher
Primary care physician supply:       Rural 59 per 100K vs.
                                       urban 100 per 100K
OB/GYN supply:                       Rural 41% of counties
                                       have no OB/GYN
────────────────────────────────────────────────────────────────
Source: CDC Rural Health, 2024; HRSA, 2024
```

Vermont's equity profile is shaped primarily by rural and economic disparities rather than racial ones (Vermont is 91% non-Hispanic white). Rural Vermonters face worse access, higher costs as a percentage of income, and worse outcomes on preventable conditions — a pattern that requires equity-centered design even in the absence of large racial disparities.

Social Determinants of Health: The Upstream Architecture of Disparities

Health disparities do not arise spontaneously. They are produced by systematically unequal exposure to the social and economic conditions that determine health. The World Health Organization defines the social determinants of health (SDOH) as "the conditions in which people are born, grow, live, work, and age."

The SDOH framework:

```
SOCIAL DETERMINANTS OF HEALTH (SDOH)
╔══════════════════════════════════════════════════════════════════╗
║  ECONOMIC STABILITY                                              ║
║  Employment, income, expenses, debt, medical bills,              ║
║  support for saving and insurance                                ║
╠══════════════════════════════════════════════════════════════════╣
║  EDUCATION ACCESS AND QUALITY                                    ║
║  Literacy, language, early childhood education,                  ║
║  vocational training, higher education                           ║
╠══════════════════════════════════════════════════════════════════╣
║  HEALTH CARE ACCESS AND QUALITY                                  ║
║  Health coverage, provider availability, quality of care,        ║
║  health literacy, language access                                ║
╠══════════════════════════════════════════════════════════════════╣
║  NEIGHBORHOOD AND BUILT ENVIRONMENT                              ║
║  Housing quality, access to healthy food, water quality,         ║
║  exposure to toxic environments, transportation access           ║
╠══════════════════════════════════════════════════════════════════╣
║  SOCIAL AND COMMUNITY CONTEXT                                    ║
║  Social cohesion, civic participation, discrimination,           ║
║  incarceration history, violence exposure                        ║
╚══════════════════════════════════════════════════════════════════╝
```

The evidence on SDOH and health:

Research consistently shows that SDOH account for 30–55% of health outcomes — more than clinical care (20–30%) and more than individual health behaviors (30%). Yet healthcare spending is almost entirely directed at clinical care. Social services spending as a share of total social spending (healthcare + social services) in the U.S. is 53% — compared to 80%+ in peer nations like Germany, France, and Sweden. Nations with higher social spending ratios consistently outperform the U.S. on health outcomes.

SDOH screening in clinical settings:

CMS and NCQA have increasingly required or incentivized SDOH screening in clinical settings, with specific ICD-10 Z-codes for social determinants:

Z-Code	Meaning
Z59.0	Homelessness
Z59.1	Inadequate housing
Z59.4	Lack of adequate food
Z59.5	Extreme poverty
Z60.2	Problems related to living alone
Z63.5	Disruption of family by separation/divorce

SDOH screening is a necessary first step, but it is not sufficient without actionable referral pathways. A clinical team that screens patients for food insecurity and has no community food resource to refer to has done nothing to address the underlying need.

The HEROI Metric: Equity-Weighted Cost-Effectiveness

Standard cost-effectiveness analysis (described in Chapter 5) treats a QALY gained by one person as equivalent to a QALY gained by any other person. This assumption has been challenged by health economists and equity advocates who argue that reducing a disparity — improving health for a person who is systematically disadvantaged — has greater social value than the same improvement for an already-advantaged person.

The **Health Equity Return on Investment (HEROI)** metric, developed and used in HTR's Health Equity Studio, applies an equity weighting to standard ICER calculations:

```
HEROI CALCULATION
────────────────────────────────────────────────────────────────
Standard ICER = ΔCost / ΔQALYs (unadjusted)

HEROI = ΔCost / (ΔQALYs × Equity Weight)

Equity Weight = 1 + (Disadvantage Score × Equity Premium Factor)

Where:
  Disadvantage Score   = composite of racial minority status (0–1),
                           income below poverty (0–1),
                           rural residence (0–1),
                           uninsured (0–1)

  Equity Premium Factor = user-selected (0.0 = no equity premium;
                           0.5 = 50% additional weight for maximum
                           disadvantage)

Example:
  Black, low-income, rural, uninsured patient
  Disadvantage Score = (1.0 + 1.0 + 1.0 + 1.0) / 4 = 1.0

  With Equity Premium Factor = 0.5:
  Equity Weight = 1 + (1.0 × 0.5) = 1.5

  Standard ICER: $80,000/QALY  →  HEROI: $53,333/QALY

Interpretation: An intervention targeting this population
delivers 50% more social value per QALY than the unadjusted
analysis suggests.
────────────────────────────────────────────────────────────────
```

HEROI does not change the clinical facts — it changes the societal value framing within which those facts are evaluated. An intervention that looks marginally cost-effective under standard analysis may be highly cost-effective under HEROI if it primarily serves disadvantaged populations.

Measuring Health Equity: From Aspirations to Metrics

NCQA's Health Equity Accreditation (HEA) Standard:

NCQA launched its Health Equity Accreditation program for health plans in 2021, establishing operational standards across six domains:

1. Data collection (race, ethnicity, language, sexual orientation, gender identity)
2. Stratified HEDIS quality measurement by demographic group
3. Quality improvement programs targeting identified disparity populations
4. Cultural and linguistic competency of the provider network
5. Community health worker deployment
6. Health literacy and patient activation programs

CMS Health Equity Index (HEI):

Beginning in 2024, CMS introduced the Health Equity Index as a component of the Medicare Advantage Star Rating system. The HEI rewards MA plans that reduce disparities in quality measure performance between their disadvantaged (dual-eligible and Low-Income Subsidy) and non-disadvantaged populations.

The AHEAD Model equity requirements:

Vermont's AHEAD Model includes explicit health equity performance expectations — a relatively unusual feature for a global budget model. Health plans and providers participating in AHEAD are expected to:

- Collect and report race/ethnicity/language data for all attributed members
- Report performance on selected quality measures stratified by race/ethnicity
- Demonstrate year-over-year improvement in identified disparity measures

HTR's assessment: these requirements are underspecified. "Equity" appears as a goal without the operational measurement framework required to know whether the model is achieving it. Vermont needs annual equity reporting tied to specific SDOH and disparity metrics, with consequences for sustained equity performance failures.

Vermont's Equity Profile: A Case Study in Non-Racial Disparities

Vermont's equity analysis illustrates that health equity is not only about racial disparities — and that predominantly white states can have significant equity challenges that require equity-centered policy design.

Rural health equity in Vermont:

Vermont is 62% rural by USDA definition — the second-highest rural population share among U.S. states (behind Maine). Rural Vermonters experience:

```
VERMONT RURAL vs. URBAN HEALTH DISPARITIES (2024)
────────────────────────────────────────────────────
                              Rural VT    Urban VT
────────────────────────────────────────────────────
Hospital distance (>30 min):  34%         6%
Primary care access wait:     23 days     11 days
Annual wellness visit rate:   62%         74%
Diabetes control (HbA1c<8):   61%         69%
Commercial premium burden:    18% income  12% income
Opioid mortality rate:        21/100K     14/100K
────────────────────────────────────────────────────
```

Economic equity:

Vermont has the second-highest commercial premium burden relative to median household income among all U.S. states. For a household earning $55,000 annually — typical for rural Vermont — a family plan costing $24,000 per year (common for small employer coverage) consumes 44% of gross income. This is not theoretical: it is the lived experience of a significant fraction of Vermont working families.

LGBTQ+ equity:

Vermont leads the nation on LGBTQ+ legal protections. But legal protections and healthcare equity are different things. LGBTQ+ Vermonters face:

- Limited access to gender-affirming care outside Burlington
- Concentration of LGBTQ+-affirming behavioral health services in one geographic area
- Higher rates of depression, anxiety, and substance use disorder — driven by social stress, not biology

HTR's recommendation: GMCB should require all licensed health plans to demonstrate LGBTQ+-affirming care network adequacy as a condition of license renewal.

The Health Equity Studio: Analytical Tools for Equity Work

HTR's Health Equity Studio provides five analytical capabilities:

1. Disparity Analysis: Performance on 10 clinical and access outcomes stratified by race/ethnicity, income quartile, and rural/urban geography. Enables identification of specific disparity populations and measures for targeted intervention.

2. Geographic Access Mapping: Drive-time analysis to nearest primary care physician, specialist, hospital, and OBGYN service by zip code. Identifies access deserts — geographic areas where populations face travel times that constitute practical access barriers.

3. SDOH Burden Scoring: County-level composite of housing instability, food insecurity, transportation barriers, and income poverty. Identifies communities with the highest social need — and therefore the highest potential return on SDOH investment.

4. HEROI Metric: Equity-weighted ICER calculation (described above) for comparing interventions that serve populations with different disadvantage profiles.

5. Equity Action Plan Generator: Ranked recommendations based on equity impact (magnitude of disparity reduction) and feasibility (availability of evidence-based interventions). Provides a prioritized roadmap for equity-centered program design.

Key Concepts Introduced in This Chapter

> **Social determinants of health (SDOH):** The conditions in which people are born, grow, live, work, and age that produce health disparities — housing, food security, income, education, transportation, social connection, environmental quality.

> **HEROI (Health Equity Return on Investment):** An equity-weighted cost-effectiveness metric that gives additional value to interventions targeting disadvantaged populations; used in HTR's Health Equity Studio.

> **Prevention Quality Indicator (PQI):** An AHRQ measure identifying hospitalizations preventable with adequate ambulatory care; used as an equity measure because rural and low-income populations have higher PQI rates.

> **NCQA Health Equity Accreditation (HEA):** NCQA's standards for health plan equity practices, covering data collection, stratified quality measurement, and disparity-focused quality improvement.

> **CMS Health Equity Index (HEI):** A Medicare Advantage Star Rating component rewarding plans that reduce quality measure disparities between disadvantaged and non-disadvantaged enrollees.

> **Racial disparity index:** A composite measure of outcome gaps between non-Hispanic white and Black and Hispanic populations across selected clinical measures; a core component of the HTI Equity sub-index.

CHAPTER 9: Operations — The Execution Test Every Reform Must Pass

The $1 Trillion Administrative Floor

The United States spends approximately $1 trillion annually on healthcare administration — billing, coding, credentialing, claims processing, prior authorization, compliance, and the enormous overhead of operating a fragmented multi-payer system. This is approximately 25–34% of total healthcare spending.

For context: if the U.S. reduced its administrative overhead to Canadian levels (~13% of spending), the savings would fund universal coverage for all currently uninsured Americans with hundreds of billions to spare.

Administrative cost in healthcare is not a minor inefficiency. It is a structural feature of the payment system — and operations is the pillar that contends with it directly.

But operations analysis is not only about reducing administrative costs. It is about the execution capacity of healthcare organizations: can they actually implement the transformation strategies that policy, economics, technology, clinical, and equity analysis prescribes? This is the question the operations pillar exists to answer.

Revenue Cycle Management: The Financial Nervous System

Revenue cycle management (RCM) is the end-to-end process of converting clinical services into revenue. In a fee-for-service world, this process runs: patient registration → insurance verification → charge capture → claim submission → adjudication → payment → denial management → collections.

In a value-based care world, RCM must evolve significantly — but it cannot abandon its fee-for-service functions simultaneously. Most health systems operate in a hybrid world where some revenue comes from FFS billing and some comes from VBC contracts. Managing both simultaneously is an operational challenge.

Key revenue cycle metrics:

```
REVENUE CYCLE BENCHMARK METRICS (2025)
────────────────────────────────────────────────────────────────
Metric                          Benchmark    Hospitals in VBC
────────────────────────────────────────────────────────────────
Clean claim submission rate      >95%         >97% (data quality req.)
Initial denial rate              <5%          <4% (attribution denials)
Days in accounts receivable      45–55 days   Depends on contract type
Net collection rate              >95%         >96% (value leakage risk)
Cost to collect (% of revenue)   <3%          Target <2.5% in VBC
Prior auth approval rate         >90%         >95% (care delay risk)
Undercoding rate (HCC RAF)        <5%          <2% (risk contract revenue)
────────────────────────────────────────────────────────────────
```

The HCC coding imperative:

Under risk-based VBC contracts (MSSP, ACO REACH, Medicare Advantage), a health system's revenue depends on the accuracy and completeness of its HCC coding. An HCC code that is clinically present but not documented and submitted to CMS results in a lower RAF score and a lower capitation payment — revenue loss that is permanent for that performance year.

RAF coding gap analysis:

```
HCC CODING GAP EXAMPLE — 500-PHYSICIAN PRACTICE
──────────────────────────────────────────────────
Attributed MA population:          12,000 patients
Expected RAF score (by claims):     1.24
Actual RAF score (coded):           1.19
RAF gap:                            0.05

Revenue impact per patient per year:
  PMPM benchmark: $1,050
  RAF adjustment: 1.19 vs. 1.24 = 4% revenue difference
  Revenue gap:    $1,050 × 12,000 × 4% = $504,000/year

Intervention: Retrospective chart review + prospective
  HCC education program
  Cost:          $85,000/year
  Revenue recovered: ~$350,000/year (partial gap closure)
  ROI:           312%
```

Prior Authorization Operations: Where Policy Meets Process

Prior authorization is both a policy issue (Chapter 4) and an operations issue. The policy question is what the rules should be. The operations question is how the health system's revenue cycle executes against those rules at scale.

Prior authorization workflow:

```
PRIOR AUTH PROCESS FLOW
──────────────────────────────────────────────────────────────────
Clinical team identifies need for service requiring PA
  ↓
Clinical documentation prepared (diagnosis, clinical indication)
  ↓
PA request submitted to payer (increasingly via electronic portal)
  ↓
Payer review: automated (AI-based) or clinical reviewer
  ↓
Decision: Approve / Deny / Pend for additional info
  ↓ (if denied)
Peer-to-peer review (physician to physician)
  ↓ (if still denied)
Formal appeal process
  ↓ (if still denied)
External independent review
──────────────────────────────────────────────────────────────────
Average time: 14 days standard; 3–5 days "urgent"
  (CMS 2024 rule: 7 calendar days standard; 72 hours urgent)
```

Electronic Prior Authorization (ePA):

The CMS 2024 Interoperability and PA Final Rule requires health plans to implement ePA by January 2026. ePA uses FHIR-based APIs to automate portions of the PA workflow — checking eligibility, submitting requests with structured clinical data, and receiving decisions electronically. This should reduce PA processing time from days to hours for straightforward cases.

Gold-carding:

Several states (Texas, West Virginia, Connecticut) have enacted "gold-carding" laws that exempt providers from PA requirements for specific services when they have a history of consistent approval. Gold-carding acknowledges that PA is a blanket administrative requirement that could be targeted more precisely — consistent high-approvers do not need individual request review.

Workforce Operations: Credentialing, Onboarding, and Retention

The operational dimensions of workforce management — as distinct from the clinical competency and policy dimensions described in Chapter 7 — are among the most undermanaged elements of healthcare operations.

Credentialing cycle time:

Provider credentialing is the process of verifying a clinician's education, training, licensure, and competence before allowing them to practice at a facility and bill for services. For a health system entering a new VBC contract that requires a significant expansion of care coordination staff, credentialing cycle time is a direct revenue and care-access constraint.

```
AVERAGE CREDENTIALING CYCLE TIMES (2025)
──────────────────────────────────────────
Primary Care Physician:    87 days
Nurse Practitioner:        72 days
Physician Assistant:       68 days
RN (nursing staff):        21 days
Community Health Worker:   35 days

If a new VBC contract starts January 1 and requires
20 new care coordinators, the first billable date
for those coordinators is approximately March 10
(72-day NP credentialing) — a $680,000 revenue gap
at $1,700/day per coordinator.
```

Workforce operational stability metrics:

The HTI Operations sub-index measures workforce operational stability through three composite indicators:

- Healthcare workforce turnover rate (RN, MD, NP, PA, clinical support)
- Clinical vacancy rate (percentage of budgeted clinical positions unfilled)
- Credentialing cycle time (average days from hire to first billable encounter)

Vermont's operational challenge:

Vermont's hospitals face compounding workforce operational challenges: high turnover rates driven by compensation competition from adjacent states (New Hampshire, Massachusetts), housing cost barriers to rural recruitment (a problem the Wyman Report correctly identifies and which policy reform addresses), and the workforce depletion effects of sustained negative operating margins — which reduce the capacity to offer competitive compensation.

Supply Chain in the Transformation Context

Healthcare supply chain management — pharmaceutical procurement, medical device and supply acquisition, GPO strategy, and facilities management — is typically treated as a back-office function. In the context of healthcare transformation, it becomes strategically important when care delivery models change.

The care migration supply chain problem:

Value-based care models systematically shift care from inpatient to outpatient, and from outpatient to community-based and home-based settings. A hospital that is successful under its global budget will reduce inpatient admissions, increase primary care visits, and expand home health and remote monitoring. Each shift changes the supply chain:

```
CARE MIGRATION SUPPLY CHAIN IMPLICATIONS
──────────────────────────────────────────────────────────────
Inpatient → Outpatient migration:
  - Reduced medical/surgical supply consumption (inpatient)
```

```
  - Increased outpatient consumables, procedure packs
  - OR utilization patterns shift
  - Linen/laundry, dietary affected

Outpatient → Home-based migration:
  - RPM devices: procurement, programming, distribution, return
  - Home health supplies: wound care, insulin, CPAP
  - Pharmacy: expanded home delivery, specialty pharmacy
  - Transportation services: patient logistics management

Key challenge: GPO contracts are designed for traditional
inpatient volume. VBC organizations need GPO strategies
aligned with their actual care delivery mix.
──────────────────────────────────────────────────────────────
```

The Administrative Overhead Paradox

The $1 trillion annual U.S. healthcare administrative cost is primarily generated by the complexity of the multi-payer payment system — not by any single organization's inefficiency. Each payer has different billing codes, different prior authorization requirements, different credentialing standards, and different quality reporting formats. The cumulative administrative burden on providers is the sum of all these differences across all their payer contracts.

The multi-payer administrative burden:

A primary care physician's practice with 3,000 patients and a typical payer mix (Medicare, Medicaid, 5–7 commercial plans) must:

- Maintain separate electronic claim formats for each payer
- Manage separate PA requirements (different services, different clinical criteria, different portals)
- Report quality measures in different formats to different plans
- Credential physicians separately with each payer
- Reconcile explanation of benefits documents from each payer's systems

A global budget model — like Vermont's AHEAD — theoretically reduces this burden by moving toward unified payment targets across payers. In practice, administrative simplification requires not only the global budget structure but the operational data infrastructure to support it.

The Revenue Cycle Friction Index

The HTI Operations sub-index measures revenue cycle friction using a composite of three national indicators:

1. Clean claim submission rate (inverse — higher rate = lower friction)
2. Average days in accounts receivable across Medicare and Medicaid
3. Claim denial rates

Revenue cycle friction and VBC readiness:

Organizations with high revenue cycle friction — high denial rates, long AR days, low clean claim rates — are less ready for VBC contracts for several reasons:

- Data quality problems that generate claim denials also generate attribution errors in VBC models
- Organizations that struggle to collect under FFS will struggle to manage the more complex financial flows of shared savings models
- Revenue cycle operational weakness signals broader data governance and process management gaps that will impede VBC reporting requirements

The HTR Hospital Financial Scorecard — one of the 19 Research Lab tools — benchmarks a health system's revenue cycle metrics against Critical Access Hospital, Rural PPS, and Urban Tertiary peer groups, enabling targeted identification of operational gaps before entering VBC contracts.

Building Operational Readiness for VBC: A Checklist

HTR's advisory practice uses the following operational readiness framework when assessing a health system's VBC transition readiness:

```
VBC OPERATIONAL READINESS ASSESSMENT FRAMEWORK
═══════════════════════════════════════════════════════════════

REVENUE CYCLE
□ HCC coding gap analysis completed and remediation underway
□ Clean claim rate >95% for Medicare and Medicaid
□ Electronic prior authorization implemented for top 20 PA services
□ Attribution data reconciliation process established
□ VBC contract financial modeling completed with CFO sign-off

WORKFORCE OPERATIONS
□ Care coordination staffing model designed and budgeted
□ Credentialing pipeline mapped; estimated cycle time documented
□ Retention risk assessment for care management workforce
□ Community health worker deployment plan developed

TECHNOLOGY (operational dimension)
□ EHR attribution list generated and validated
□ Care gap identification workflow embedded in clinical workflows
□ Quality measure reporting automated (no manual data pulls)
□ SDOH screening tool implemented in registration workflow

COMPLIANCE AND ACCREDITATION
□ HIPAA data sharing agreements in place for all VBC partners
□ Joint Commission compliance review completed
□ NCQA PCMH recognition in place for participating practices
□ CMS Conditions of Participation compliance verified

SUPPLY CHAIN
□ GPO strategy reviewed for alignment with projected care mix shift
□ Home health supply procurement partnerships established
□ RPM device procurement, programming, and return workflow documented
═══════════════════════════════════════════════════════════════
```

Key Concepts Introduced in This Chapter

> **Revenue cycle management (RCM):** The end-to-end process of converting clinical services into revenue, from patient registration through claim submission, adjudication, and collections.

> **Clean claim rate:** The percentage of claims submitted to payers that are accepted and processed without error on the first submission; a key revenue cycle quality metric.

> **Days in accounts receivable (AR):** The average number of days between service delivery and payment collection; a core revenue cycle efficiency metric.

> **RAF coding gap:** The difference between a patient population's expected RAF score (based on clinical records) and their submitted RAF score (based on coded claims); revenue loss from undercoding in risk-bearing contracts.

> **Gold-carding:** State legislation or payer policy exempting consistently high-approving providers from prior authorization requirements for specific services.

> **Credentialing cycle time:** The number of days from a new provider's hire date to their first billable clinical encounter; a key workforce operations metric with direct revenue implications.

> **Revenue Cycle Friction Index:** HTR's composite measure of revenue cycle operational inefficiency, combining clean claim rates, AR days, and denial rates; a component of the Operations sub-index.

End of Part II
